VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 5 | JEE 2022 (Gen 1 & 2)

Date: 30/11/2020 Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

Candidate's Signature : _ _ _ _ _ _ _ _ _ _ **Invigilator's Signature** _ _ _ _ _ _ _ _

PART I : PHYSICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The acceleration due to gravity at the surface of a spherical planet of radius R and uniform density ρ is proportional to:

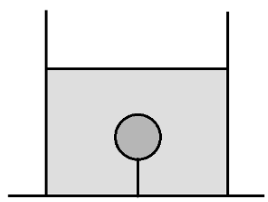
(A) ρR **(B)** ρR^{-1} **(C)** $\rho^{-1} R^{-2}$ **(D)** ρR^{2}

2. Starting from rest, a uniform circular disc rolls without slipping down a plane of inclination θ. The initial height of the centre of the disc above the ground is h. The time taken by the disc to reach the bottom of the plane is:

(A) proportional to $\frac{1}{\tan\theta}$ **(B)** proportional to $\frac{1}{\sin\theta}$

(C) proportional to $\frac{1}{\sqrt{\sin\theta}}$ **(D)** independent of θ

3. An object of mass m is submerged completely inside water filled in a container kept on a horizontal surface as shown. The object is tied to the bottom of the vessel by a massless, thin thread. In equilibrium, the tension in the thread is T. The relative density of the object is:

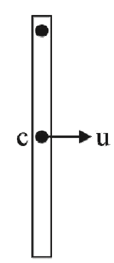


(A) $\frac{mg-T}{mg}$ **(B)** $\frac{T}{mg}$ **(C)** $\frac{mg}{T}$ **(D)** $\frac{mg}{mg+T}$

4. The time period of a satellite in a circular orbit around the earth is T. The kinetic energy of the satellite is proportional to T^{-n}. Then, n is equal to:

(A) $\frac{1}{2}$ **(B)** $\frac{2}{3}$ **(C)** $\frac{4}{3}$ **(D)** $\frac{3}{2}$

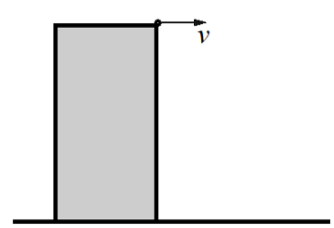
5. A thin uniform rod of mass m and length ℓ is hinged at its top end and hangs vertically. Now its centre is given a horizontal velocity u as shown. What is the minimum value of u so that the rod can loop around in a vertical circle?



(A) $\sqrt{4g\ell}$ **(B)** $\sqrt{2.5g\ell}$

(C) $\sqrt{2g\ell}$ **(D)** $\sqrt{1.5g\ell}$

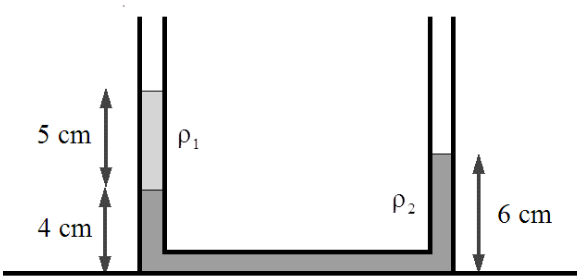
6. A particle of mass m is projected horizontally with velocity v from the top of a building of height h. Just before it strikes the horizontal ground, the angular momentum of the particle about the point of projection is:



(A) $2mvh$ **(B)** $\sqrt{3}mvh$

(C) mvh **(D)** $\frac{1}{\sqrt{2}}mvh$

7. A narrow U-tube placed on a horizontal surface contains two liquids of densities ρ_1 and ρ_2. The columns of the liquids are indicated in the figure. The ratio $\frac{\rho_1}{\rho_2}$ is equal to:



(A) $\frac{1}{3}$ **(B)** $\frac{2}{5}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{4}{5}$

8. Two water pipes of diameters 2 cm and 4 cm are connected in series i.e end to end. The velocity of flow of water in the pipe of 2 cm diameter is

(A) 4 times that in the other pipe **(B)** $\frac{1}{4}$ times that in the other pipe

(C) 2 times that in the other pipe **(D)** $\frac{1}{2}$ times that in the other pipe

9. An object is projected from the surface of the earth at an angle θ with the surface. The object escapes to infinity. Assume that the only force acting on the object is the gravitational force due to the earth. During its journey, the angular momentum of the object about the centre of the earth:

(A) decreases continuously **(B)** increases continuously
(C) increases initially, then decreases **(D)** remains constant

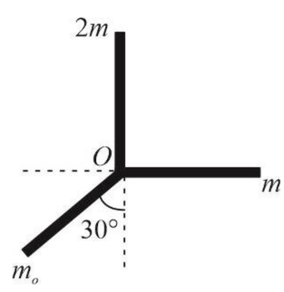
10. Three thin, uniform rods of the same length and mass m_0, $2m$ and m are joined to form a composite body with all three rods in the same plane, and this body is pivoted at the junction point O. The body remains in equilibrium with the rods of mass $2m$ and m being vertical and horizontal respectively, as shown. All three rods are in the vertical plane. The correct relation between m_0 and m is:

(A) $m_0 = \frac{\sqrt{3}}{2}m$

(B) $m_0 = 2m$

(C) $m_0 = \frac{2}{\sqrt{3}}m$

(D) $m_0 = \frac{m}{2}$



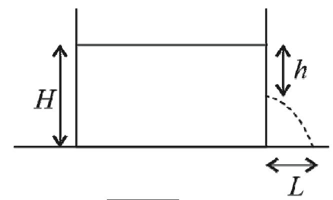
11. The retarding torque on a ceiling fan due to air resistance is proportional to $\omega^{3/2}$, where ω is the angular velocity of the fan. The electric power needed to keep the fan rotating at a constant angular velocity ω is proportional to:

(A) $\omega^{3/2}$ **(B)** ω^2 **(C)** $\omega^{5/2}$ **(D)** ω^3

12. The acceleration due to gravity at the surface of the earth is g. The acceleration due to gravity at a height $\frac{1}{100}$ times the radius of the earth above the surface is close to:

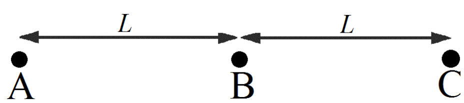
(A) $\left(\frac{96}{100}\right)g$ **(B)** $\left(\frac{98}{100}\right)g$ **(C)** $\left(\frac{99}{100}\right)g$ **(D)** $\left(\frac{101}{100}\right)g$

13. A liquid is filled in a container to a height H. A small hole is opened in the side of the container a height h below the surface of the liquid. The distance L from the bottom of the container of the point where the stream of liquid leaving the container meets the ground is:



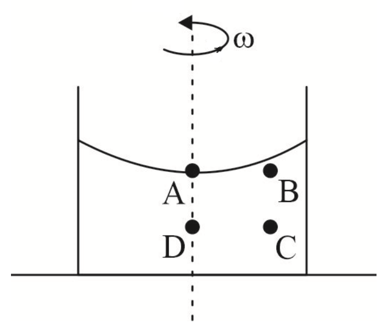
(A) $2\sqrt{h(H-h)}$ **(B)** $\sqrt{h(H-h)}$ **(C)** $2\sqrt{\frac{h^2}{H-h}}$ **(D)** $\sqrt{\frac{h^2}{H-h}}$

14. Three particles A, B and C of mass m, m and $2m$ lie in a straight line as shown. The only force acting on these particles is due to the gravitational attraction towards each other. Let the magnitude of net force on them be F_A, F_B and F_C respectively. Then, $F_A : F_B : F_C$ is equal to:



(A) 3 : 2 : 5 **(B)** 3 : 1 : 4 **(C)** 2 : 2 : 3 **(D)** 6 : 1 : 5

15. A cylindrical container filled with a liquid is being rotated about its central axis at a constant angular velocity ω. Four points A, B, C and D are chosen in the same plane such that ABCD is a square of side length a and AB is horizontal while BC is vertical. A and D lie on the axis of rotation. Let the pressure at A, B, C and D be denoted by P_A, P_B, P_C and P_D. Which of these options is correct?



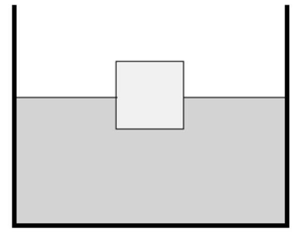
(A) $P_C > P_A$ for all values of ω; $P_B < P_D$ only if $\omega > \sqrt{\frac{2g}{a}}$

(B) $P_C > P_A$ and $P_B < P_D$ only if $\omega > \sqrt{\frac{2g}{a}}$

(C) $P_C > P_A$ for all values of ω; $P_B > P_D$ only if $\omega > \sqrt{\frac{2g}{a}}$

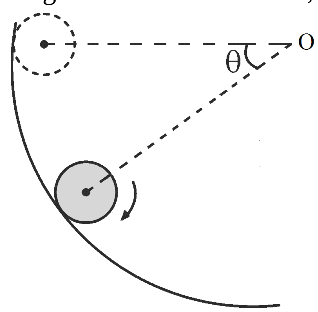
(D) $P_C > P_A$ and $P_B > P_D$ for all values of ω

16. A solid cube floats in a liquid of density ρ with half of its volume submerged as shown. The side length of the cube is a, which is very small compared to the size of the container. Now a liquid of density $\frac{\rho}{3}$ (immiscible with the liquid already in the container) is poured slowly into the container. The column of the liquid of density $\frac{\rho}{3}$ that must be poured so that the cube is fully submerged (with its top surface coincident with the surface of the poured liquid) is:



(A) $\frac{a}{3}$ **(B)** $\frac{a}{2}$ **(C)** $\frac{2a}{3}$ **(D)** $\frac{3a}{4}$

17. A solid sphere of radius r starts rolling down a circular track (centred at O) from rest, such that the centre of the sphere moves in a circular path of radius R. In its initial position, the line joining the centre of the sphere to O is horizontal. The sphere always rolls without slipping. At the instant the line joining the centre of the sphere to O makes an angle θ with the horizontal, the angular velocity of the sphere is:

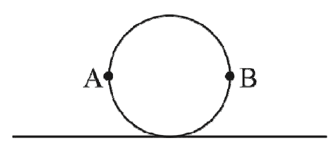


(A) $\sqrt{\frac{10gR(1-\cos\theta)}{7r^2}}$ **(B)** $\sqrt{\frac{10gR\sin\theta}{7r^2}}$ **(C)** $\sqrt{\frac{5gR(1-\cos\theta)}{7r^2}}$ **(D)** $\sqrt{\frac{5gR\cos\theta}{7r^2}}$

18. A particle is given a velocity $\frac{v_e}{\sqrt{3}}$ in a vertically upward direction from the surface of the earth, where v_e is the escape velocity from the surface of the earth. Let the radius of the earth be R. The height of the particle above the surface of the earth at the instant it comes to rest is:

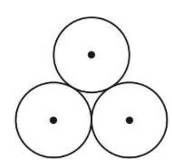
(A) $3R$ **(B)** $2R$ **(C)** $\frac{3R}{4}$ **(D)** $\frac{R}{2}$

19. A circular disc rolls without slipping on a horizontal surface such that the velocity of its centre is v. At a particular instant, let A and B be the leftmost and rightmost points on the disc. Let $\bar{v}_{AB}$ be the velocity of A relative to B at this instant. Then, the magnitude of $\bar{v}_{AB}$ is:



(A) $2v$ **(B)** v **(C)** $2\sqrt{2}v$ **(D)** $\sqrt{2}v$

20. Three identical solid spheres of mass m and radius R each are joined together to form a rigid body such that their centres form an equilateral triangle. The moment of inertia of this body about an axis passing through the centres of two of the spheres is:



(A) $\frac{19}{5}mR^2$ **(B)** $\frac{21}{5}mR^2$ **(C)** $\frac{31}{20}mR^2$ **(D)** $\frac{39}{20}mR^2$

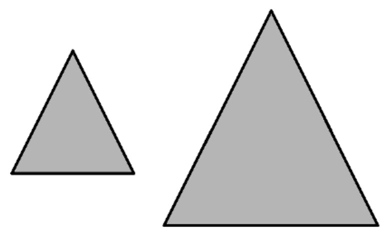
SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

21. A small wooden block of density $400\ kg/m^3$ is released from rest inside a swimming pool filled with water of density $10^3 kg/m^3$. The acceleration of the block just after it is released is ________ m/s^2. $(g = 10 m/s^2)$

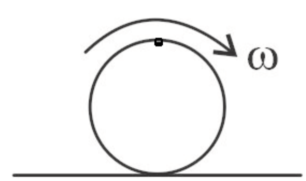
22. Two equilateral triangles of side lengths a and $2a$ respectively are cut out from a large thin, uniform metallic sheet. The moment of inertia of the first triangle about one of its sides is I_1 and the moment of inertia of the second triangle about one of its sides is I_2. The ratio $\frac{I_2}{I_1}$ is equal to _____________.



23. A sphere is made of an alloy of Metal A (density $8\ g/cm^3$) and Metal B (density $6\ g/cm^3$). The sphere floats in mercury (density $13.6\ g/cm^3$) with half its volume submerged. The percentage of the total volume of the sphere that is occupied by metal A is ___________ .

24. A cylindrical tube with a frictionless piston ends in a small hole of area of cross–section $2 \times 10^{-6} m^2$, which is very small as compared to the area of cross-section of the tube. The tube is filled with a liquid of density $10^4\ kg/m^3$. The tube is held horizontal and the piston is pushed at a constant speed so that liquid comes out of the hole at a constant rate $'Q'\ m^3/s$. No liquid leaks around the piston. The material of the tube can bear a maximum pressure 6×10^5 Pa. The maximum possible value of Q so that the tube does not break is __________ $\times 10^{-5} m^3/s$. [Atmospheric pressure $= 10^5 Pa$]

25. A small particle of mass m is fixed rigidly to the rim of a uniform circular ring of mass m and radius R. The composite body is made to roll without slipping on a horizontal surface. At a particular instant, the particle is at the top most point in its trajectory and the angular velocity of the composite body is ω. The kinetic energy of the body at this instant is $X(mR^2\omega^2)$. The value of X is __________ .



SPACE FOR ROUGH WORK

PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which of the following set of thermodynamic parameters is valid for expansion of the Universe ?

(A) $\Delta G_{sys} < 0; \Delta S_{sys} < 0$ **(B)** $\Delta G_{sys} > 0; \Delta S_{sys} > 0$

(C) $\Delta G_{sys} = 0; \Delta S_{surr} < 0$ **(D)** $\Delta G_{sys} < 0; \Delta S_{sys} > 0$

2. For which of the following reaction; yield of the product/s will increase by addition of an inert gas at constant pressure ?

(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ **(B)** $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(C) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ **(D)** $C(\text{solid, diamond}) \rightleftharpoons C(\text{solid, graphite})$

3. Select correct statement out of following :

(A) $\Delta_f H^\circ_{H_2O(\ell)}$ is zero at 298 K and 1 atm

(B) Entropy is zero for all substances at the zero Kelvin

(C) $\Delta_f H^\circ_{H^+(aq)}$ is taken as zero according to the IUPAC convention

(D) $\Delta_f H^\circ_{H(g)}$ is zero at 298 K and 1 atm

4. Ice is in equilibrium with water at certain temperature and pressure. If applied pressure over the system is increased then :

(A) Amount of water at equilibrium will increase

(B) Amount of ice at equilibrium will increase

(C) Amount of water at equilibrium remain unchanged

(D) Melting point of the ice will increase

5. Which of the following statement is NOT correct for an adiabatic process involving an ideal gas ?

(A) More cooling of the gas during its reversible expansion as compared to its irreversible expansion

(B) More heating of the gas during its reversible compression as compared to its irreversible compression

(C) Gas will expand by consuming its internal energy

(D) There will be NO heat exchange between the system and its surrounding

6. For which of the following exothermic process the heat liberated at constant pressure will be less than the heat liberated at constant volume ?

(A) $2C(s) + O_2(g) \longrightarrow 2CO(g)$ **(B)** $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

(C) $H_2(g) + I_2(g) \longrightarrow 2HI(g)$ **(D)** $C(s) + O_2(g) \longrightarrow CO_2(g)$

7. For the reaction $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$, $K_p = 3.5 \times 10^{-3}$ atm at 27°C. Value of K_c at this temperature will be : $[R = 0.0821 \text{ L-atm/mole-Kelvin}]$

(A) 1.42×10^{-4} **(B)** 1.42×10^{-3} **(C)** 1.42×10^{-5} **(D)** 1.42×10^{-6}

8. Select correct statement out of the following :

(A) Equilibrium between reactant and products inside a closed vessel is NOT dynamic in nature.

(B) Addition of some amount of radioactive sugar to a saturated solution of non-radioactive sugar shows radioactivity in solid phase only after sometime

(C) The equilibrium $CO_2(g) \rightleftharpoons CO_2(aq)$ can be disturbed by change in temperature as well as pressure

(D) For all reversible reactions active mass of the reactant is equal to active mass of the product at the equilibrium state.

9. Given that

Standard molar enthalpy of atomization of graphite $= 720\,kJ/mole$

Standard molar enthalpy of atomization of $Cl_2(g) = 240\,kJ/mole$

Standard molar enthalpy of vaporization of $CCl_4(\ell) = 30\,kJ/mole$

Standard molar enthalpy of formation of $CCl_4(\ell) = -130\,kJ/mole$

Then enthalpy of the following reaction will be : [in kJ]

$$CCl_4(g) \longrightarrow C(g) + 4Cl(g)$$

(A) 3310 **(B)** 2310 **(C)** 1520 **(D)** 1300

10. Work done by an ideal gas will be maximum for the thermodynamic process

$(P_1, V_1, T_1) \longrightarrow (P_2, V_2, T_2)$ if :

(A) $P_1 = P_2;\ V_2 > V_1$ and $T_2 > T_1$

(B) $T_1 = T_2;\ P_1 > P_2$ and $V_2 > V_1$

(C) $V_1 = V_2;\ P_1 < P_2$ and $T_1 < T_2$

(D) $P_1V_1^{\gamma} = P_2V_2^{\gamma};\ V_2 > V_1,\ \gamma = C_P/C_V$ (where C_P and C_V are molar heat capacity of the gas at constant pressure and constant volume respectively)

11. Certain quantity of an ideal gas undergo expansion from 5L to 10L against a constant external pressure equal to 2 atm at 300 Kelvin, then for the process :

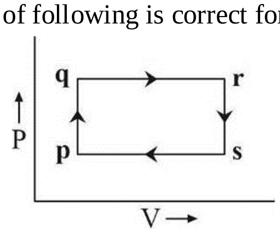
(A) Change in internal energy, $\Delta U = 0$ for the gas

(B) Heat exchange, $Q = 0$ for the gas

(C) Mechanical work, $W = 0$ for the gas

(D) Entropy change, $\Delta S = 0$ for the gas

12. Which of following is correct for on ideal gas system involved in the given reversible cyclic process ?



(A) Adiabatic heating $(p \to q)$ **(B)** Isobaric heating $(q \to r)$

(C) Isochoric heating $(r \to s)$ **(D)** Isoentropic expansion $(s \to p)$

13. Given that

For $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ $K_p = 10^{-1}$ atm at 125°C

For $2NO(g) + O_2 \rightleftharpoons 2NO_2(g)$ $K_p = 10^{-2}\ atm^{-1}$ at 125°C

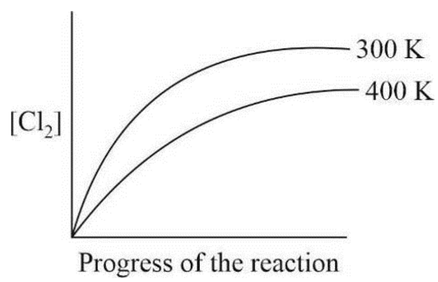
And for $2NO_2(g) \rightleftharpoons N_2O_4(g)$ $K_p = 5\,atm^{-1}$ at 125°C

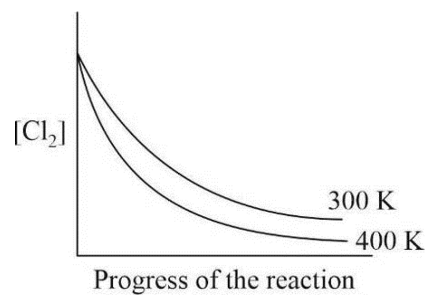
Then K_p for $N_2O_4(g) \rightleftharpoons N_2(g) + 2O_2(g)$ at 125°C is :

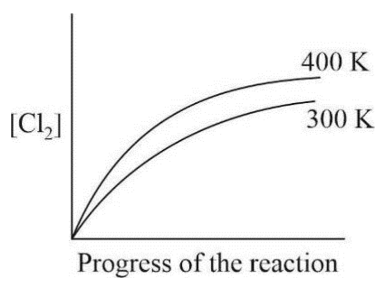
(A) 20 atm **(B)** 200 atm **(C)** 400 atm **(D)** 100 atm

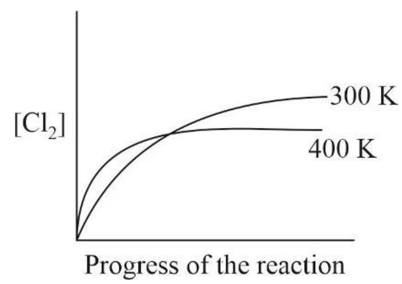
14. Which of the following graphical representation is correct for the reversible chemical reaction ?

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) 

(B) 

(C) 

(D) 

15. Thermal decomposition of $(NH_4)_2CO_3(s)$ was carried out in a closed vessel, starting with 0.2 atm pressure of $NH_3(g)$. If at equilibrium total pressure of the gases is 1 atm then K_p(in atm^4) of the following reaction will be :

$(NH_4)_2CO_3(s) \rightleftharpoons 2NH_3(g) + H_2O(g) + CO_2(g)$

(A) 1.44×10^{-2} **(B)** 1.44×10^{-3} **(C)** 1.57×10^{-3} **(D)** 1.57×10^{-4}

16. Which of the following process is entropy driven ?

(A) Condensation **(B)** Freezing **(C)** Vaporization **(D)** Crystallization

17. For which of following thermodynamic process entropy change of the system (ideal gas) is showing different values under reversible and irreversible conditions ?

(A) Isothermal process **(B)** Isobaric process

(C) Isochoric process **(D)** Adiabatic process

18. As ionization energy is defined at absolute zero and ionization enthalpy at any other temperature. If ionization energy for a metal M is E_0 then its ionization enthalpy at temperature T is :

(Given C_p for monoatomic gaseous substance is $\frac{5}{2}R$)

(A) $E_0 - \frac{3}{2}RT$ **(B)** $E_0 + \frac{3}{2}RT$ **(C)** $E_0 + \frac{5}{2}RT$ **(D)** $E_0 - \frac{5}{2}RT$

19. An ideal gas "X" undergo reversible adiabatic expansion from T_i to T_f. To perform maximum work, "X" should be :

(A) $H_2(g)$ **(B)** $Ne(g)$ **(C)** $CO_2(g)$ **(D)** $CH_4(g)$

20. Which of the following reaction is nearest to completion at 300 K ?

(A) $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g);\ K_c = 5\times10^{18}$ at 300 K

(B) $Cl_2(g) + F_2(g) \rightleftharpoons 2ClF(g);\ K_c = 4\times10^{-5}$ at 300 K

(C) $CO_2(g) + C(s) \rightleftharpoons 2CO(g);\ K_c = 0.2$ at 300 K

(D) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g);\ K_c = 3.5\times10^{-10}$ at 300 K

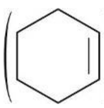
SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. $NO_2(g)$ got dimerise by 80% at certain temperature T. Vapour density of the equilibrium mixture is________. [Atomic mass: Nitrogen = 14, Oxygen = 16]

22. The change in entropy when two moles of a monoatomic perfect gas is compressed to half its volume and simultaneously heated to twice its initial temperature is ________. [in cal/K]
$[\ln 2 = 0.69,\ R = 2\,\text{cal/mole-K}]$

23. Hydrogenation enthalpy of cyclohexene () is $-119\,\text{kJ/mole}$ while that of benzene (C_6H_6) is $-208\,\text{kJ/mole}$, then resonance energy of benzene is ________. [kJ/mole]

24. What will be value of equilibrium constant for the reaction

$AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2Cl(aq)$ at 300 K if given that :

$$\Delta_f G^\circ_{(AgCl(s))} = -20\,\text{kJ/mole}$$

$$\Delta_f G^\circ_{(NH_3(aq))} = -10\,\text{kJ/mole}$$

$$\Delta_f G^\circ_{(Ag(NH_3)_2Cl(aq))} = -40\,\text{kJ/mole}$$

25. 300 J of energy is supplied to 3 moles of an ideal gas at constant pressure such that its temperature is increased by 2.5 Kelvin. Value of molar heat capacity of the gas at constant volume is ________.
$(\text{J/mole-K})\ [R = 8.3\,\text{J/mole-K}]$

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The square of the length of tangent from $(2,-3)$ on the circle $x^2+y^2-2x-4y-4=0$ is :

(A) 15 **(B)** 17 **(C)** 20 **(D)** 30

2. If the two circles $(x-2)^2+(y+3)^2=\lambda^2$ and $x^2+y^2-4x+4y-1=0$ intersect in two distinct points then:

(A) $\lambda<-2$ **(B)** $-2<\lambda<4$ **(C)** $2<\lambda<4$ **(D)** $\lambda=-2$

3. The equations of the normal to the circle $x^2+y^2-8x-2y+12=0$ at the points whose ordinate is -1 will be:

(A) $2x-y-7=0, 2x+y-9=0$ **(B)** $2x+y-7=0, 2x+y+9=0$

(C) $2x+y+7=0, 2x+y+9=0$ **(D)** None of these

4. The line $x=c$ cuts the triangle with corners (0, 0); (1, 1) and (9, 1) into two regions. For the area of the two regions to be the same c must be equal to :

(A) 5/2 **(B)** 3 only **(C)** 7/2 **(D)** 3 or 15

5. If the second term of the expansion $\left[a^{1/13}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{5/2}$ then the value of $\frac{{}^nC_3}{{}^nC_2}$ is :

(A) 4 **(B)** 3 **(C)** 12 **(D)** 6

6. The number of values of `r' satisfying the equation, ${}^{39}C_{3r-1}-{}^{39}C_{r^2}={}^{39}C_{r^2-1}-{}^{39}C_{3r}$ is :

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

7. If the lines $x+y+3=0$ and $2x-y-9=0$ lie along diameters of a circle of circumference 8π then the equation of the circle is :

(A) $x^2+y^2-4x+10y+13=0$ **(B)** $x^2+y^2+4x-10y+13=0$

(C) $x^2+y^2-4x-10y+13=0$ **(D)** $x^2+y^2+4x+10y+13=0$

8. The point diametrically opposite to the point $(6,0)$ on the circle $x^2+y^2-4x+6y-12=0$ is :

(A) $(2,6)$ **(B)** $(-2,6)$ **(C)** $(2,-6)$ **(D)** $(-2,-6)$

9. The co-ordinate of vertices A, B and C of ΔABC *are* $(1,-2),(-7,6)$ and $\left(\frac{11}{5},\frac{2}{5}\right)$ respectively.

The measure of the interior angle A of the ΔABC, is :

(A) acute and lies in (75°, 90°) **(B)** acute and lies in (60°, 75°)

(C) acute and lies in (45°, 60°) **(D)** obtuse and lies in (120°, 150°)

10. The remainder when $4^{87}+6^{87}$ is divided by 25, is :

(A) 19 **(B)** 20 **(C)** 21 **(D)** 22

11. The largest real value for x such that $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{4-k!}\right)\left(\frac{x^k}{k!}\right)=\frac{8}{3}$ is :

(A) $2\sqrt{2}-5$ **(B)** $2\sqrt{2}+5$ **(C)** $-2\sqrt{5}-5$ **(D)** $-2\sqrt{2}+5$

12. Consider a family of lines $(4a+3)x-(a+1)y-(2a+1)=0$ where $a \in R$. A member of this family with positive gradient making an angle of $\pi/4$ with the line $3x-4y=2$, is :

(A) $7x-y-5=0$ **(B)** $4x-3y+2=0$ **(C)** $x+7y=15$ **(D)** $5x-3y-4=0$

13. The line $y=mx+c$ will be a normal to the circle with radius r and centre (a,b), if :

(A) $a=mb+c$ **(B)** $b=ma+c$ **(C)** $c=ma+b$ **(D)** $c=mb+a$

14. The two circles $x^2+y^2-cx=0$ and $x^2+y^2=4$ touch each other if :

(A) $|c|=4$ **(B)** $|c|=2$ **(C)** $|c|=8$ **(D)** None of these

15. If $(1+x+x^2)^{25}=a_0+a_1x+a_2x^2+....+a_{50}\cdot x^{50}$ then $a_0+a_2+a_4+....+a_{50}$ is :

(A) even **(B)** odd and of the form $3n$

(C) odd and of the form $(3n-1)$ **(D)** odd and of the form $(3n+1)$

16. The coefficient of x^{11} in the expression $(1+x)^5\,(3+x)^4\,(7+x)^3$ equals :

(A) 28 **(B)** 34 **(C)** 38 **(D)** 42

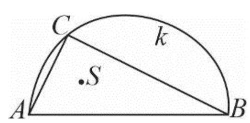
17. Given $(1-2x+5x^2-10x^3)\,(1+x)^n=1+a_1x+a_2x^2+....$ and that $a_1^2=2a_2$ then the value of n is :

(A) 6 **(B)** 2 **(C)** 5 **(D)** 3

18. The two circles $x^2+y^2+2ax+c=0$ and $x^2+y^2+2by+c=0$ touch if $\frac{1}{a^2}+\frac{1}{b^2}=$

(A) $1/c$ **(B)** c **(C)** $\frac{1}{c^2}$ **(D)** c^2

19. AB is the diameter of a semicircle k, C is an arbitrary point on the semicircle (other than A or B) and S is the centre of the circle inscribed in the triangle ABC, then measure of :



(A) angle ASB changes as C moves on k

(B) angle ASB is the same for all positions of C but it cannot be determined without knowing the radius

(C) angle $ASB=135°$ for all positions of C

(D) angle $ASB=150°$ for all positions of C

20. A point which is inside the circle $x^2+y^2+3x-3y+2=0$ is :

(A) $(-1,3)$ **(B)** $(-2,1)$ **(C)** $(2,1)$ **(D)** None of these

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. If $\sum_{k=0}^{100}\left(\frac{k}{k+1}\right){}^{100}C_k = \frac{a(2^{100})+b}{c}$ where $a, b, c \in N$, then find the least value of $(a+b+c)$.

22. The number of common tangents to the circle $x^2+y^2-2x-4y-4=0$ and $x^2+y^2+4x+8y-5=0$ is_________.

23. $(1+x)(1+x+x^2)(1+x+x^2+x^3).....(1+x+x^2+....+x^{100})$ when written in the ascending power of x then the highest exponent of x is _______.

24. In a circle with centre $`O'$ PA and PB are two chords. PC is the chord that bisects the angle APB. The tangent to the circle at C is drawn meeting PA and PB extended at Q and R respectively. If $QC=3$, $QA=2$ and $RC=4$, then length of RB equals a/b where a and b are co-prime then $a+b$ is_______.

25. In a triangle ABC, if $A(2,-1)$ and $7x-10y+1=0$ and $3x-2y+5=0$ are equations of an altitude and an angle bisector respectively drawn from B, then equation of BC is $ax+by+17=0$ then $a-b$ is______.

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